Ewen Gueguen

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Education

McGill University April 2028 (Expected)
B.Eng in Software Engineering (Co-op) (GPA: **3.94**) *Montreal, QC, Canada*

- **Honors (2250$)**: Faculty of Engineering Scholarship, Women Associates Scholarship in Engineering
- **Relevant Coursework**: Data Structure and Algorithms, Model-Based Programming, Software Systems

Experience

National Bank Capital Markets May 2026 – Aug 2026
Quantitative / Software Developer Intern *Montreal, QC, Canada*

- Designed a distributed inference and training platform for trader-facing quantitative models using **Ray** and **FastAPI**
- Scaled model execution to 1000+ req/s via low-latency serving and dynamic routing across isolated environments
- Segmented 192 bond firms across 353K RFQs into 4 behavioral tiers using **KMeans/GMM** with BIC selection
- Engineered 30+ RFQ features that neutralize competition, notional and sample bias to isolate true client quality
- Validated tiers with **PCA** projections and achieved 0.93 quadratic-weighted kappa stability across weight schemes

Cerence AI Jan 2026 – May 2026
Software Developer Intern *Montreal, QC, Canada*

- Integrated **Python** function tooling in an LLM routing pipeline, cutting latency by 70% and inference costs by 30%
- Enhanced LLM routing accuracy and efficiency by 35%+ through optimized categorization and tool selection rules
- Built scalable benchmarking and data validation pipelines, automating 20,000+ test cases for system reliability
- Built a deployment pipeline with **Python** and **GitLab CI/CD**, speeding up release tagging and versioning by 85%

NDT Technologies Inc. May 2025 – Aug 2025
Software Developer Intern *Montreal, QC, Canada*

- Enabled automated analysis of 10,000+ test results per day by developing **C++** pipeline linking tests to **MySQL**
- Built data visualization **PHP** web pages with dynamic filtering, pagination, and search, handling 15M+ records
- Cut real-time laser calibration errors by 35% by adding a **C++** outlier-filtering feature to line-detection software
- Modernized legacy **C/C++** application with GTK3-to-GTK4 migration, improving performance and maintainability

NDT Technologies Inc. Jun 2024 – Aug 2024
Software Developer Intern *Montreal, QC, Canada*

- Replaced a subscription-based tool and improved accounting data search efficiency by over 60% by designing and developing an internal **C++** data visualization tool, integrated with a **MySQL** database
- Reduced manual labor by 93% with **PowerShell** scripts for automating file search, organization and manipulation
- Developed reusable **SQL** queries to automate MySQL data workflows, improving query performance by 200%

Projects

OmniClaw | *Python, FastAPI, MCP, React, Web Scraping (Playwright, BeautifulSoup), API Integration* GitHub

- Engineered an **MCP** server paired with an AI orchestrator to extract and serve legacy portal student data
- Developed a cross-platform conversational AI (Web, Discord, CLI) unifying fragmented academic data

Internify | *Next.js, TypeScript, Browserbase, Stagehand, Bun, GPT* GitHub

- Designed a **Next.js** web application automating job search and submission, reducing manual work by 90%
- Architected a GPT-5-mini agent that parses and autofills multi-step job forms (<5% error) with Browserbase

LeagueOfStudies | *Next.js, TypeScript, Supabase, PostgreSQL, Gemini* GitHub

- Built a real-time multiplayer learning app with **Next.js** and **PostgreSQL**, for interactive studying with competition
- Developed an **AI content pipeline** using Gemini 2.5 Flash transforming PDFs/text into structured study questions

Technical Skills

Programming Languages: Python, Java, C/C++, TypeScript/JavaScript, C#, OCaml, PHP
Frameworks: React, Next.js, Vue, Vite, TailwindCSS, Flutter
Backend / Databases: FastAPI, Ray Core/Serve, Spring Boot, .NET, MySQL, PostgreSQL, Redis, MongoDB, Supabase
Tools / Platforms: Git, CI/CD, AWS, Linux, MCP, TCP/IP, Docker, Celery, Gradle, Cucumber